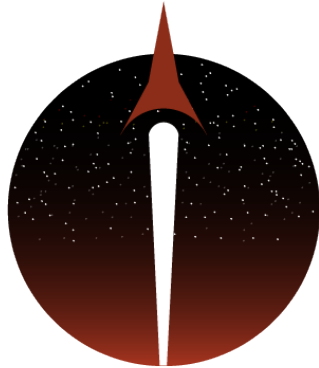




Porto Space Team
FEUP Universidade do Porto

Propulsion Department R&D

- Igniter Production -

Porto Space Team

May 28, 2025

Contents

1	Introduction	1
2	Initial Research	1
2.1	Previous Igniter - INVICTUS 1.0	1
2.2	Improvements	1
2.2.1	Sorbitol Stability	3
2.2.2	Erythritol Stability	3
APPENDIX		6
Appendix A		6
2.2.3	Chemical Nature and Mode of Action	6
2.2.4	Experimental Evidence of Effectiveness	6
3	Production Protocol	1
3.1	Materials & Safety Equipment	2
3.2	Step 1: Preparation	3
3.3	Step 2: Mixing	4
3.4	Step 3: Casting	4
3.5	Step 4: Post Processing	4
4	Testing Protocol	1
4.1	Thermocouple-Based Heat Measurement of Igniters	2

1 Introduction

This document is the property of the Porto Space Team, which retains all rights and privileges associated with its content. It is intended exclusively for use within the Propulsion Department and is subject to the team's internal guidelines and confidentiality policies.

The purpose of this document is to serve as a comprehensive record of the research and development process behind the development and production of Igniters.

Igniter is a crucial component of the INVICTUS 2.0's motor, it is meant to heat the fuel grain in order to grant combustion.

2 Initial Research

2.1 Previous Igniter - INVICTUS 1.0

Last year's method of ignition was a solid igniter which sat on top of the grain and was activated with e-matches. The chemical composition followed a simple 'rocket candy' recipe as shown in table 1.

Reagent	Sorbitol (C ₆ H ₁₄ O ₆)	Potassium Nitrate (KNO ₃)
wt %	65	35

Table 1: INVICTUS 1.0's Igniter Chemical Composition

The Igniter had a simple doughnut shape with no casing to maximize the contact with the fuel grain, avoiding the ports, as seen in figure1. Although it functioned, it had its downsides. For starters, the mixer is highly hydrophilic, requiring specialized storage, and refusing to ignite with e-matches if it absorbs too much humidity. Secondly, the chemical composition is not optimized, leaving room for higher heat fluxes to the surface.



Figure 1: Invictus 1.0 igniters

2.2 Improvements

In order to improve the hydrophilicity and storage, there are 2 solutions: either changing the chemical composition; or covering the igniter with a protective casing. At Euroc24, one of the judges mentioned the lower hydrophilicity of Erythritol when compared to Sorbitol (around 10 times), the downside be-

ing cost. A common design choice among Euroc teams is a 3D printed casing covering the igniter mixture.

When it comes to heat flux, some of the additives utilized by Euroc teams are **oxides** (e.g., Fe_2O_3) which lower reaction kinetics, increasing the burn time, **metal charge** (e.g., Mg powder) which increases the heat transfer and reaction temperature, and **reactive reagents** such as gunpowder which might lower the activation energy for the reaction.

Oxley et al. [5] studied a plethora of oxidizer-fuel mixtures. When comparing erythritol to other sugars/sweeteners, it possessed the highest heat release, 2438 J/g. The ignition temperature of the mixture is 350 °C (623 K) as seen in figure 2.

To improve processing stability, antioxidant agents such as Irganox 1010 can be added, as explored in annex 2.2.2. It also extends shelf life in addition to facilitate processing.

		KIO ₄	KMnO ₄	KBrO ₃	KClO ₃	K ₂ Cr ₂ O ₇	AP	KClO ₄	KIO ₃	KNO ₃	AN	KNO ₂
Oxidizer phase change (endo)		543	[240]	415	358	402	248	307, 613		132,331	128,167	45, 419
Oxidizer decomposition (exo)		330	277	428	474		359	630	555	703	254	510
	Endo											
Sucrose	185, 238	148	194	179	167	224	263	443	159	372	142	204
Pentaerythritol	190, 263, 359			211	209					420	242	
Erythritol	122, 297	119	112		192	383	258	313, 593	141	350	260	261
Sulfur	107, 119, 135, 179, 316	182	189	193	149		391	428	169	294	172	189
Cyclododecanol	78	185, 251	196	379	397	321	323		414	NR<500		
Cyclododecanol		334					264					
Naphthalene	80	255	265	391	397	420	356		600	600	279	
Benzoic acid 20%	121	259	123	355	189	381	393	395	403	NR<500	270	NR<500
Charcoal		344	277		300	397	353	470	436	409	203	346
Al (20%)	660	830?	750	850?	364		902	563	830?	747	846	679
Al		339	280	419			280		581		291	

Figure 2: Observed departure from baseline, Oxley et al.

The properties of the reagents can be found in table 2.,

Property	Erythritol	Sorbitol	Potassium Nitrate
Chemical Formula	$C_4H_{10}O_6$	$C_6H_{14}O_6$	KNO_3
Melting Temperature ($^{\circ}C$)	121	95	334
Boiling Temperature ($^{\circ}C$)	329	296	400
Density (g/cm ³)	1.45	1.49	2.11
CAS Number	149-32-6	50-70-4	7757-79-1
NFPA 704			

Table 2: Properties of the different reagents

2.2.1 Sorbitol Stability

Thermogravimetric analysis (TGA) indicates that sorbitol remains thermally stable up to 200–220 $^{\circ}C$, beyond which decomposition reactions initiate. These reactions include dehydration and fragmentation, leading to mass loss and volatile degradation products. The maximum rate of decomposition typically occurs around 250–270 $^{\circ}C$, with significant degradation completed by 300–350 $^{\circ}C$, depending on the heating rate and environment [1], [2].

In oxidative conditions (e.g., air), the degradation involves free radical and oxidative pathways, accelerating browning and decomposition. Under inert atmospheres, sorbitol demonstrates improved thermal resilience, suggesting that oxygen plays a catalytic role in initiating decomposition [2].

One of the most notable visual indicators of thermal degradation in sorbitol is browning, which occurs upon prolonged heating. Unlike simple caramelization, this browning is associated with the formation of conjugated carbonyl compounds and polymeric degradation products, which also impact mechanical performance and reactivity [4]. Discoloration and increased viscosity of the melt during heating are warning signs of incipient degradation.

2.2.2 Erythritol Stability

Erythritol undergoes two primary degradation processes: oxidation and dehydration, both of which lead to a loss of performance and physical changes in the material. Under an oxidative atmosphere (air), oxidation initiates through hydrogen abstraction either from the carbon backbone or hydroxyl groups. These reactions result in the formation of alkyl and alkoxy radicals, which subsequently generate carbonyl and fragmentation products. The process promotes increased volatility and, notably, sample browning, a key visual indicator of

degradation. Dehydration leads to the formation of unsaturated carbonyl compounds such as enones, which have been implicated in the discoloration and browning of various organic materials. Importantly, erythritol is not prone to caramelization or Maillard-type browning; instead, the browning observed during degradation is due to the accumulation of conjugated carbonyl structures [3].

In the context of propellant formulations—particularly those involving potassium nitrate and erythritol (commonly referred to as KNE)—these findings underscore the importance of tightly controlling processing conditions:

- Thermal treatment during casting should remain below 130°C to avoid initiating degradation.
- Exposure to oxygen should be minimized through the use of inert atmospheres or vacuum-sealed processing.
- The addition of stabilizers or antioxidants can prolong the usable life of the formulation, although their depletion must be considered in storage protocols.
- Discoloration, stickiness, or delayed solidification are warning signs of thermal degradation that should trigger batch rejection or reprocessing.

References

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APPENDIX

Appendix A

2.2.3 Chemical Nature and Mode of Action

Irganox 1010 (CAS No. 6683-19-8), chemically known as Pentaerythritol tetrakis(3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate), is a sterically hindered phenol. Its large, bulky side groups provide thermal stability and steric protection to the phenolic hydroxyl group, which is responsible for its radical-scavenging properties.

Its mechanism of action centers around radical chain-breaking. When thermally stressed erythritol undergoes oxidation, it forms free radicals such as:

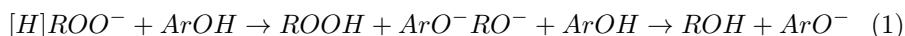
Alkyl radicals ($R^\cdot$)

Alkoxy radicals ($RO^\cdot$)

Peroxy radicals ($ROO^\cdot$)

These radicals can propagate degradation reactions, leading to molecular fragmentation and the formation of conjugated carbonyl structures responsible for discoloration and loss of crystallinity.

Irganox 1010 interrupts this process by donating a hydrogen atom to neutralize these radicals:



The resulting phenoxy radical ($ArO^\cdot$) is relatively stable due to delocalization across the aromatic ring, and can be further quenched or recombine to form non-radical products. In some cases, the phenolic antioxidant can be regenerated through secondary reactions, extending its effectiveness during thermal treatment.

2.2.4 Experimental Evidence of Effectiveness

In the referenced study, erythritol samples doped with 5 wt.% Irganox 1010 exhibited:

- Significantly lower mass loss rates under air at 121–141 °C.
- Reduced latent heat degradation, with rates of 0.13–0.15 kJ/(kg·h), compared to up to 0.58 kJ/(kg·h) in untreated erythritol.

- Improved color retention, avoiding the intense browning seen in untreated samples.
- Enhanced recrystallization behavior, with samples retaining solid phase behavior even after long-term heating (935 h) in air.

The antioxidant's performance was especially notable up to 600 hours in long-term tests, after which its effectiveness diminished, likely due to consumption or thermal exhaustion of the active compound [3].

3 Production Protocol



Igniter Production Protocol

- Propulsion Department -

Porto Space Team

Approved: _____

Propulsion Department Leader

Approved: _____

Adviser Reviewer

Approved: _____

Professor Reviewer

May 28, 2025

3.1 Materials & Safety Equipment

In order to ensure the safety of personel in the laboratory, the following Personal Protection Equipment (PPE) are required:

- Lab-coat
- Adequate laboratory attire (long trousers, impermeable shoes / boots)
- Nitrile gloves

List of equipment required:

- Fume hood
- Heating plate
- Precision balance
- Metal beakers
- Steel scoops/spatulas
- 3D-printed ASA molds
- Steel mixing rods (if possible avoid glass rods)

List of Consumables:

- Potassium nitrate (KNO_3)
- Erythritol ($\text{C}_4\text{H}_{10}\text{O}_6$)
- Sorbitol ($\text{C}_6\text{H}_{14}\text{O}_6$)
- Silicone grease
- Tracing paper
- Aluminum foil

3.2 Step 1: Preparation

Determine the mass of reactants needed:

$$m_{comp} = \frac{w_{comp} * \rho_{comp} * V_{mold}}{w_{KNO3} * \rho_{fuel} + w_{fuel} * \rho_{KNO3}} \quad (2)$$

Add 10-15% excess to account for spillage/evaporation. Leftover mixture can be reused if uncontaminated. Mixture ratio (mass): 65% KNO_3 and 35% mix of sorbitol and erythritol.

Notes:

3.3 Step 2: Mixing

Combine reactants in a metal beaker and dry-mix thoroughly.

Heat under a fume hood to 135°C while stirring vigorously with a steel rod.

- Critical: Avoid caramelization by maintaining 135°C and continuous stirring.
- Avoid using glass rods: Stress causes breakage.

.

Optional: Insert e-match at the surface of the mixture before it solidifies. If placed too soon, it can shatter.

3.4 Step 3: Casting

Pre-heat and glaze the molds using silicone grease for easier un-molding. Next, pour molten mixture into the molds using high-temperature gloves.

- Tap molds gently to release air bubbles.
- Caution: Molten mixture can cause burns; avoid direct skin contact.

Let cool for 30-40 minutes until fully solidified.

- PLA molds are unsuitable (melting point < 135°C).

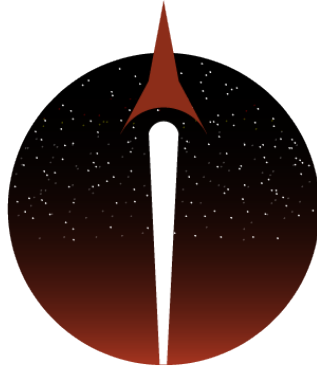
3.5 Step 4: Post Processing

Remove disks

Wrap disks in aluminum foil to prevent moisture absorption.

Store in airtight containers with desiccants (<30% humidity).

4 Testing Protocol



Porto Space Team
FEUP Universidade do Porto

Igniter Testing Protocol

Porto Space Team

Approved: _____
Propulsion Department Leader

Approved: _____
Avionics Department Leader

Approved: _____
Professor Reviewer

Approved: _____
Pyromen - Propulsion Department member

Approved: _____
Electronics Technician - Avionics Department member

May 28, 2025

4.1 Thermocouple-Based Heat Measurement of Igniters

To estimate the heat output of igniters over time, we will employ a thermocouple-based setup. The aim is not to obtain exact temperature values but to capture the peak temperature and general temperature-time profile. This makes thermocouples a cost-effective and practical choice for our purposes.

The Avionics Department has provided a metal plate equipped with two thermocouples, which we will use for testing. As the thermocouple tips are plastic-coated, we will remove this coating to expose the platinum elements, which are more suitable for high-temperature measurements.

Test Setup and Procedure:

- Two igniters—one using sorbitol and the other erythritol—will be mounted on a stable, non-flammable surface to ensure safety and consistent positioning.
- Thermocouples will be placed close to or lightly touching the igniter material to capture the temperature response accurately.
- The tests will be conducted in a safe, well-ventilated area (location to be determined). Ignition will be initiated manually using matches, and temperature data will be recorded throughout the process.

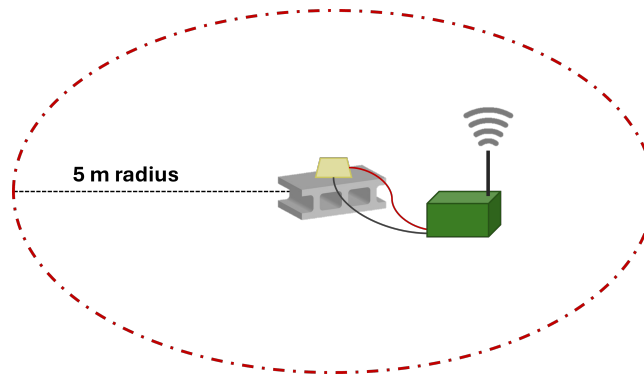


Figure 3: Test site illustration with 5 m safety radius.

• Safety Precautions:

- ☐ Tests will be conducted outdoors or in a designated, fire-safe area.
- ☐ Fire extinguisher, sand bucket, and first-aid kit will be available on-site.
- ☐ Personnel must wear safety goggles, heat-resistant gloves, non-synthetic long sleeves, and lab coats.
- ☐ Only one operator will approach the rig to ignite the igniter, while others maintain a safe distance.
- ☐ Emergency procedures will be reviewed with all team members before starting the test.

• Equipment

- ☐ Safety Equipment
- ☐ Thermocouples.
- ☐ Igniters securely stored.
- ☐ First-aid kit.
- ☐ All personnel equipped (PPE): goggles, gloves, long sleeves, lab coat.
- ☐ Ignition method.
- ☐ Non-flammable

. Testing procedure:

- ☐ Emergency procedures reviewed with the team, **safety briefing**.
- ☐ First-aid kit accessible at the test site.
- ☐ All personnel equipped with **PPE**: goggles, gloves, long sleeves.
- ☐ Fire extinguisher and sand bucket available and functional.
- ☐ Electronics and ignition equipment checked and functional:
 - ☐ Ignition channel must be unarmed.
 - ☐ Thermocouples inspected and plastic coating removed.
 - ☐ Thermocouples reading conformation.
 - ☐ Fixing E-matches to igniter.
 - ☐ Igniters securely mounted on a stable, non-flammable base.
 - ☐ Everyone **must** keep safe distance marked (**5 m**) except by **Pyromen**.
 - ☐ Pyromen arms ignition channel and confirms readiness.
 - ☐ **Everyone** must safe distance marked (**5 m**).
- ☐ Countdown for ignition.
- ☐ After ignition wait for 2 minutes before approaching the Test site.
- ☐ **Pyromen** disarms ignition channel.
- ☐ Repeat every stage from "Electronics and ignition equipment checked and functional:" for each sample tested.

Risk Assessment Table:

Potential Hazard	Consequence	Mitigation Measure
Plastic on thermocouple tips not removed	Melting, false readings	Manual removal and visual inspection before test
Igniter dislodgement during burn	Uncontrolled flame or movement	Use clamps or heat-resistant adhesive to secure
Operator injury during ignition	Burns or eye injury	Only one trained operator approaches; full PPE mandatory
Fire spread due to misfire or casing failure	Fire damage or personal injury	Test conducted outdoors; fire extinguisher and sand bucket nearby
Exposure to fumes or combustion byproducts	Respiratory irritation	Perform test in open, ventilated space
Poor emergency coordination	Delayed response and increased risk	Pre-test briefing and assigned safety roles